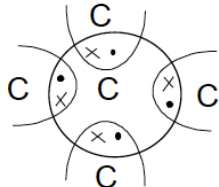


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Covalent bonding (1) Each carbon atom {shares electron pairs/bonds} with four other carbon atoms (1)  (1)	3			3	4	
		(ii)		Each carbon atom is joined to three others	1			1		
		(iii)		In graphite, the fourth electron is {free / delocalised/not involved in bonding} (1) and able to move (1) In diamond, there are no free electrons (1)	3			3		
		(iv)		In graphite, the carbon atoms are arranged in layers joined by weak forces (1) This allows (each layer to) slide over the other (easily) (1) In diamond, there (is a rigid structure/no layers) held together by strong (covalent) bonds (1)	3			3		
	(b)			Any 3 × (1) from: Carbon nanotubes have higher strength / stronger (1) The lower density / lighter (1) They are much better conductors (1) There can be a greater distance between pylons / greater breaking length (1) Accept converse Comments about other properties are neutral		3		3		
				Question 6 total	10	3		13		